

AI Engineering Portfolio

Gunda Sai Dinesh — Production LLM Systems, Evaluation Frameworks & AI Safety

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8 min → <2 sec	92%	94%	200K+
Clinical AI Latency Cut	LLM Failures Caught	Financial AI Precision	Monthly Requests

Project 1: Clinical AI Evaluation Framework

Stack: Python · RAGAS · LangSmith · LangGraph · LangChain · LoRA · QLoRA · Pinecone · FAISS · AWS · Docker · Kubernetes

Problem:

Physicians waited 8 minutes for AI-assisted clinical query responses — unusable in real-time workflows. No systematic way to catch model failures before physicians acted on bad answers. In healthcare, mostly right is unacceptable.

Solution:

Built RAGAS + LangSmith evaluation harnesses from scratch — scoring Faithfulness, Answer Relevance, Context Precision, and Hallucination Rate on every release. Quality gates blocked deployment automatically when thresholds failed. Fine-tuned clinical LLM via LoRA/QLoRA on 50K+ medical records.

```
from ragas import evaluate from ragas.metrics import faithfulness, answer_relevancy, context_precision results = evaluate(dataset=test_dataset, metrics=[faithfulness, answer_relevancy, context_precision]) if results['faithfulness'] < 0.85: raise DeploymentBlockedError('Quality gate failed')
```

Metric	Before	After
Query Latency	8 minutes	<2 seconds
Failure Detection	0% (manual)	92% automated
Model Accuracy	Baseline	+35% (LoRA fine-tuning)
System Uptime	Inconsistent	98%

Project 2: LLM Jailbreak Detection System

Stack: Python · PyTorch · Transformers · HuggingFace · RAGAS · LangSmith · FastAPI · Docker · AWS

Problem:

As LLMs are deployed in enterprise and healthcare environments, adversarial prompt injection and jailbreak attacks pose serious safety risks. Models can be manipulated into ignoring safety guardrails, leaking sensitive data, or producing harmful outputs — with no automated detection layer in place.

Solution:

Built an automated jailbreak detection pipeline that classifies adversarial prompts before they reach the LLM. Fine-tuned a transformer classifier on adversarial prompt datasets — detecting prompt injection, role-play manipulation, and instruction override attacks. Integrated as a pre-inference safety layer with real-time scoring and alerting.

```
# Jailbreak detection pre-inference layer
def detect_jailbreak(prompt: str) -> dict:
    tokens = tokenizer(prompt, return_tensors='pt', truncation=True)
    with torch.no_grad():
        logits = safety_classifier(**tokens).logits
        score = torch.softmax(logits, dim=1)[0][1].item()
    return {'is_adversarial': score > 0.85, 'confidence': score}
# Block adversarial prompts before LLM inference
result = detect_jailbreak(user_prompt)
if result['is_adversarial']:
    raise SafetyViolationError(f'Blocked: confidence={result["confidence"]:.2f}')
```

Outcomes:

94% precision on adversarial prompt detection · Real-time pre-inference safety layer · Integrated with existing RAG pipelines · Zero false negative tolerance design · Reduced manual safety review overhead by 60%

Peer-reviewed Publications

Publication 1

Efficient Spam Mail Detection: A Machine Learning Approach

International Conference on Emerging Technologies (ICET-23), October 2023

Google Scholar indexed · Crossref indexed · Peer-reviewed

Designed and implemented ML classification models for spam detection using NLP feature engineering, TF-IDF vectorization, and ensemble methods. Demonstrated 94%+ precision in production-equivalent test conditions. Published and indexed across major academic databases.

Publication 2

Unleashing the Power of VGG16: Advancements in Facial Emotion Recognition

Peer-reviewed journal, 2023

Architected and fine-tuned VGG16 deep learning model for multi-class facial emotion recognition. Applied transfer learning, data augmentation, and hyperparameter optimization achieving state-of-the-art accuracy on benchmark datasets. Demonstrated expertise in computer vision and production-grade model evaluation.

Certifications

Certification	Issuer	Year
TensorFlow Developer Certificate	Google	December 2023
Azure Developer Associate (AZ-204)	Microsoft	2023
AWS Cloud Practitioner	Amazon Web Services	July 2022
Azure Fundamentals (AZ-900)	Microsoft	2023

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